Name……..Signature…….

P525/2
CHEMISTRY
(Paper 2)
APRIL
2026
Duration; 2hrs

MATEETE COMPREHESIVE SEED SECONDARY SCHOOL

END OF TERM ONE ASSESSMENT 2026

Uganda advanced certificate of education

CHEMISTRY
Practical

SENIOR FIVE

2hours

INSTRUCTIONS TO LEARNERS

This paper consists of only one item and it is ***compulsory***

Use blue or black ink. Any work done in pencil except drawings will not be scored

Write your responses in ***spaces provided****. No additional sheets of paper should be inserted on this paper.*

Mathematical tables and silent non programmable scientific calculators may be used

Reference books (ie textbooks, booklets) etc. should not be used

You are not allowed *to start working with the apparatus for the* ***first 10 minutes****. This time is to enable you to read the item and make sure that you have all the apparatus and chemicals you require*

For examiner's use only						
CODE						**TOTAL**
SCORE						

ITEM

Success Detergents Limited is a company that deals in the manufacture of soaps and detergents. The company uses anhydrous sodium sulphate as a filler in powdered detergens. The anhydrous sodium sulphate is initially prepared from a reaction between sodium hydrogencarbonate and dilute sulphuric acid. The senior industrial chemist of the company guided that the stock solution of sodium sulphate must be prepared from a sodium hydrogencarbonate solution whose molarity is greater than **0.1M** and dilute sulphuric acid whose concentration is less than **0.5M**.

The newly recruited industrial chemist of the company has been tasked to establish whether the stock solution of sulphuric acid in FA1 which is present in the company stores is suitable for the reaction with sodium hydrogen carbonate solution during the preparation of sodium sulphate.

You have been contacted to assist the newly recruited member of the technical staff UNBS and you have been provided with the following solutions from the company store with other apparatus and materials;

130cm^{3}of FA1 which is a solution of dilute sulphuric acid, **3.1g of solid K** which is sodium hydrogencarbonate ($NaHCO_3$), 2 beakers, measuring cylinder, 250cm^{3}volumetric flask, burette, pipette, conical flask, retort stand with clamp, methyl orange indicator and access to distilled water.

TASK

As a chemistry student, design and carry out a scientific investigation to guide the newly recrited member of staff of UNBS on what to do.

Take atomic masses: C = 12; H = 1 and O = 16 Na=23

Your design and investigation should include the following:

a) **Aim, hypothesis variables, apparatus/requirements, materials, procedure, Risks and mitigation**

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b) Results (a minimum of three readings is required)

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c) **Analysis of your results and determination of the molar concentration of the dilute sulphuric acid present in the company's store**

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d) Conclusion and advise/recommendations

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